



National Load Despatch Centre
राष्ट्रीय भार प्रेषण केंद्र
GRID CONTROLLER OF INDIA LIMITED
ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड

(Government of India Enterprise/ भारत सरकार का उद्यम)
B-9, QUTUB INSTITUTIONAL AREA, KATWARIA SARAI, NEW DELHI -110016
बी-9, कुतुब इंस्टीट्यूशनल एरिया, कटवारिया सराये, न्यू दिल्ली-110016

Ref: POSOCO/NLDC/SO/Daily PSP Report

दिनांक: 28th April 2024

To,

- कार्यकारी निदेशक, पू.क्षे.भा.प्रे.के., 14, गोल्फ क्लब रोड, कोलकाता - 700033
Executive Director, ERLDC, 14 Golf Club Road, Tollygunge, Kolkata, 700033
- कार्यकारी निदेशक, ऊ.क्षे.भा.प्रे.के., 18/ए, शाहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली - 110016
Executive Director, NRLDC, 18-A, Shaheed Jeet Singh Marg, Katwaria Sarai, New Delhi - 110016
- कार्यकारी निदेशक, प.क्षे.भा.प्रे.के., एफ3, एम आई डी सी क्षेत्र, अंधेरी, मुंबई -400093
Executive Director, WRLDC, F-3, M.I.D.C. Area, Marol, Andheri (East), Mumbai-400093
- कार्यकारी निदेशक, ऊ.पू.क्षे.भा.प्रे.के., डोंगतेह, लोअर नोंग्रह, लापालंग, शिलोंग - 793006
Executive Director, NERLDC, Dongteih, Lower Nongrah, Lapalang, Shillong - 793006, Meghalaya
- कार्यकारी निदेशक, द.क्षे.भा.प्रे.के., 29, रेस कोर्स क्रॉस रोड, बंगलुरु -560009
Executive Director, SRLDC, 29, Race Course Cross Road, Bangalore-560009

Sub: Daily PSP Report for the date 27.04.2024.

महोदय/Dear Sir,

आईईजीसी-2010 की धारा स.-5.5.1 के प्रावधान के अनुसार, दिनांक 27-अप्रैल-2024 की अखिल भारतीय प्रणाली की दैनिक ग्रिड निष्पादन रिपोर्ट रांभांप्रेके की वेबसाइट पर उपलब्ध है।

As per article 5.5.1 of the Indian Electricity Grid Code, the daily report pertaining power supply position of All India Power System for the date 27th April 2024, is available at the NLDC website.

धन्यवाद,

Report for previous day

Date of Reporting: 28-Apr-2024

A. Power Supply Position at All India and Regional level

	NR	WR	SR	ER	NER	TOTAL
Demand Met during Evening Peak hrs(MW) (at 20:00 hrs; from RLDCs)	62001	65382	53625	28098	2876	211982
Peak Shortage (MW)	0	812	0	1142	0	1954
Energy Met (MU)	1352	1602	1354	652	56	5016
Hydro Gen (MU)	190	62	63	28	11	354
Wind Gen (MU)	19	77	38	-	-	134
Solar Gen (MU)*	168.44	91.73	140.23	3.80	1.07	405
Energy Shortage (MU)	1.07	6.11	0.95	6.98	0.00	15.11
Maximum Demand Met During the Day (MW) (From NLDC SCADA)	62385	73620	64064	30480	3279	222968
Time Of Maximum Demand Met	19:48	15:40	15:23	00:02	18:49	15:49

B. Frequency Profile (%)

Region	FVI	< 49.7	49.7 - 49.8	49.8 - 49.9	< 49.9	49.9 - 50.05	> 50.05
All India	0.029	0.00	0.91	4.33	5.24	85.80	8.96

C. Power Supply Position in States

Region	States	Max.Demand Met during the day (MW)	Shortage during maximum Demand (MW)	Energy Met (MU)	Drawal Schedule (MU)	OD(+)/UD(-) (MU)	Max OD (MW)	Energy Shortage (MU)
NR	Punjab	8301	0	163.3	55.9	-2.2	222	0.00
	Haryana	9386	0	172.2	97.0	-1.1	139	0.00
	Rajasthan	13707	0	280.4	59.6	2.9	581	0.91
	Delhi	4963	0	106.1	87.5	0.3	182	0.00
	UP	24247	0	496.8	207.1	2.7	867	0.00
	Uttarakhand	2167	0	48.0	26.1	0.6	277	0.00
	HP	1578	0	31.3	13.3	-0.5	88	0.16
	J&K(UT) & Ladakh(UT)	2283	0	45.5	30.6	-0.9	71	0.00
	Chandigarh	235	0	4.8	4.9	-0.1	22	0.00
WR	Railways_NR ISTS	191	0	3.8	3.4	0.4	66	0.00
	Chhattisgarh	6062	0	137.1	74.6	0.2	341	0.00
	Gujarat	23547	0	481.8	163.7	-2.4	466	0.00
	MP	12549	0	277.8	165.7	0.3	854	5.23
	Maharashtra	28331	0	627.4	183.2	-0.8	1016	0.88
	Goa	765	0	15.7	11.2	4.3	1	0.00
	DNHDDPDCL	1292	0	29.6	29.1	0.5	165	0.00
	AMNSIL	879	0	19.9	9.3	0.0	273	0.00
	BALCO	524	0	12.5	12.5	0.0	7	0.00
SR	Andhra Pradesh	12951	0	250.8	77.0	-1.3	436	0.00
	Telangana	11032	0	227.8	96.9	1.0	517	0.00
	Karnataka	16181	0	319.4	143.9	-1.6	1106	0.00
	Kerala	5563	450	110.1	83.7	2.5	610	0.95
	Tamil Nadu	20237	0	435.1	271.1	-0.7	577	0.00
	Puducherry	496	0	11.1	10.6	-0.1	30	0.00
ER	Bihar	6725	0	139.8	128.2	5.6	738	5.81
	DVC	3492	0	73.6	-42.6	-1.7	225	0.00
	Jharkhand	2004	0	45.8	30.9	1.0	233	1.17
	Odisha	6356	0	132.6	51.3	1.3	510	0.00
	West Bengal	12254	0	258.2	106.9	-1.2	302	0.00
	Sikkim	87	0	1.4	1.4	0.0	45	0.00
	Railways_ER ISTS	14	0	0.1	0.1	0.0	4	0.00
NER	Arunachal Pradesh	185	0	2.8	2.5	0.1	59	0.00
	Assam	2012	0	35.6	29.5	0.0	221	0.00
	Manipur	148	0	2.5	2.5	-0.1	35	0.00
	Meghalaya	279	0	5.4	3.7	-0.1	87	0.00
	Mizoram	128	0	1.7	1.9	-0.4	21	0.00
	Nagaland	147	0	2.3	2.3	-0.1	19	0.00
	Tripura	335	0	5.8	5.8	0.4	91	0.00

D. Transnational Exchanges (MU) - Import(+ve)/Export(-ve)

	Bhutan	Nepal	Bangladesh	Godda -> Bangladesh
Actual (MU)	-4.8	-9.5	-23.1	-29.7
Day Peak (MW)	-352.0	-777.1	-1094.0	-1513.0

E. Import/Export by Regions (in MU) - Import(+ve)/Export(-ve); OD(+)/UD(-)

	NR	WR	SR	ER	NER	TOTAL
Schedule(MU)	114.1	-313.1	202.2	-13.2	10.0	0.0
Actual(MU)	107.6	-312.2	205.3	-10.9	9.7	-0.5
O/D/U/D(MU)	-6.5	0.9	3.1	2.3	-0.3	-0.5

F. Generation Outage(MW)

	NR	WR	SR	ER	NER	TOTAL	% Share
Central Sector	2840	8520	3018	4086	351	18815	53
State Sector	6256	6727	1855	1954	129	16920	47
Total	9095	15246	4873	6040	480	35734	100

G. Sourcewise generation (Gross) (MU)

	NR	WR	SR	ER	NER	All India	% Share
Coal	851	1672	808	716	15	4062	75
Lignite	25	17	66	0	0	108	2
Hydro	190	62	63	28	11	354	7
Nuclear	25	50	75	0	0	151	3
Gas, Naptha & Diesel	47	94	7	0	25	173	3
RES (Wind, Solar, Biomass & Others)	199	172	211	5	1	588	11
Total	1337	2068	1230	749	52	5436	100

Share of RES in total generation (%)	14.88	8.33	17.11	0.62	2.06	10.81
Share of Non-fossil fuel (Hydro,Nuclear and RES) in total generation(%)	30.96	13.77	28.36	4.41	23.20	20.10

H. All India Demand Diversity Factor

Based on Regional Max Demands	1.048
Based on State Max Demands	1.083

I. All India Peak Demand and shortage at Solar and Non-Solar Hour

	Max Demand Met(MW)	Time	Shortage(MW)
Solar hr	222968	15:49	469
Non-Solar hr	218674	22:41	3401

Diversity factor = Sum of regional or state maximum demands / All India maximum demand

**Note: All generation MU figures are gross

***Godda (Jharkhand) -> Bangladesh power exchange is through the radial connection (isolated from Indian Grid)

Solar Hours -> 07:00 to 17:00hrs and rest are Non-Solar Hours

*Source: RLDCs for solar connected to ISTS; SLDCs for embedded solar. Limited visibility of embedded solar data.

Executive Director-NLDC

INTER-REGIONAL EXCHANGES

Import=(+ve) /Export =(-ve) for NET (MU)

Date of Reporting: 28-Apr-2024

Sl No	Voltage Level	Line Details	No. of Circuit	Max Import (MW)	Max Export (MW)	Import (MU)	Export (MU)	NET (MU)
Import/Export of ER (With NR)								
1	HVDC	ALIPURDUAR-AGRA	2	0	0	0.0	0.0	0.0
2	HVDC	PUSAULI B/B	-	0	48	0.0	1.1	-1.1
3	765 kV	GAYA-VARANASI	2	1194	133	7.4	0.0	7.4
4	765 kV	SASARAM-FATEHPUR	1	413	246	0.0	1.6	-1.6
5	765 kV	GAYA-BALIA	1	34	431	0.0	6.2	-6.2
6	400 kV	PUSAULI-VARANASI	1	0	133	0.0	1.7	-1.7
7	400 kV	PUSAULI -ALLAHABAD	1	85	14	0.6	0.0	0.6
8	400 kV	MUZAFFARPUR-GORAKHPUR	2	818	154	5.3	0.0	5.3
9	400 kV	PATNA-BALIA	2	548	517	0.0	4.2	-4.2
10	400 kV	NAUBATPUR-BALIA	2	230	149	0.0	0.6	-0.6
11	400 kV	BIHARSHARIFF-BALIA	2	639	0	7.0	0.0	7.0
12	400 kV	MOTIHARI-GORAKHPUR	2	427	268	0.0	1.0	-1.0
13	400 kV	BIHARSHARIFF-VARANASI	2	134	174	0.0	0.0	0.0
14	220 kV	SAHUPURI-KARAMNASA	1	84	81	0.0	0.4	-0.4
15	132 kV	NAGAR UNTARI-RIHAND	1	0	0	0.0	0.0	0.0
16	132 kV	GARWAH-RIHAND	1	30	0	0.8	0.0	0.8
17	132 kV	KARMANASA-SAHUPURI	1	0	0	0.0	0.0	0.0
18	132 kV	KARMANASA-CHANDAULI	1	0	0	0.0	0.0	0.0
ER-NR						21.2	16.8	4.4
Import/Export of ER (With WR)								
1	765 kV	JHARSUGUDA-DHARAMJAIGARH	4	1437	0	21.7	0.0	21.7
2	765 kV	NEW RANCHI-DHARAMJAIGARH	2	1303	0	21.1	0.0	21.1
3	765 kV	JHARSUGUDA-DURG	2	0	480	0.0	7.0	-7.0
4	400 kV	JHARSUGUDA-RAIGARH	4	0	444	0.0	12.3	-12.3
5	400 kV	RANCHI-SIPAT	2	257	26	3.0	0.0	3.0
6	220 kV	BUDHIPADAR-RAIGARH	1	35	88	0.0	0.7	-0.7
7	220 kV	BUDHIPADAR-KORBA	2	82	35	0.5	0.0	0.5
ER-WR						46.2	20.1	26.2
Import/Export of ER (With SR)								
1	HVDC	JEYPORE-GAZUWAKA B/B	2	0	521	0.0	12.1	-12.1
2	HVDC	TALCHER-KOLAR BIPOLE	2	0	1497	0.0	36.2	-36.2
3	765 kV	ANGUL-SRIKAKULAM	2	0	2407	0.0	44.1	-44.1
4	400 kV	TALCHER-I/C	2	417	0	8.3	0.0	8.3
5	220 kV	BALIMELA-UPPER-SILERRU	1	0	0	0.0	0.0	0.0
ER-SR						0.0	92.3	-92.3
Import/Export of ER (With NER)								
1	400 kV	BINAGURI-BONGAIGAON	2	352	176	2.4	0.8	1.6
2	400 kV	ALIPURDUAR-BONGAIGAON	2	617	155	4.4	0.0	4.4
3	220 kV	ALIPURDUAR-SALAKATI	2	87	37	0.5	0.0	0.5
ER-NER						7.3	0.8	6.5
Import/Export of NER (With NR)								
1	HVDC	BISWANATH CHARIALI-AGRA	2	658	0	15.6	0.0	15.6
NER-NR						15.6	0.0	15.6
Import/Export of WR (With NR)								
1	HVDC	CHAMPA-KURUKSHETRA	2	0	3011	0.0	70.1	-70.1
2	HVDC	VINDHYACHAL B/B	-	491	249	3.9	1.3	2.6
3	HVDC	MUNDRA-MOHINDERGARH	2	0	1445	0.0	36.4	-36.4
4	765 kV	GWALIOR-AGRA	2	141	1408	0.1	17.2	-17.1
5	765 kV	GWALIOR-PHAGI	2	1392	1034	6.8	8.2	-1.4
6	765 kV	JABALPUR-ORAI	2	396	551	0.0	9.1	-9.1
7	765 kV	GWALIOR-ORAI	1	517	0	7.0	0.0	7.0
8	765 kV	SATNA-ORAI	1	0	821	0.0	15.5	-15.5
9	765 kV	BANASKANTHA-CHITORGARH	2	1950	46	18.4	0.0	18.4
10	765 kV	VINDHYACHAL-VARANASI	2	0	2889	0.0	54.2	-54.2
11	400 kV	ZERDA-KANKROLI	1	415	0	4.5	0.0	4.5
12	400 kV	ZERDA -BHINMAL	1	414	0	4.5	0.0	4.5
13	400 kV	VINDHYACHAL -RIHAND	1	971	0	22.8	0.0	22.8
14	400 kV	RAPP-SHUJALPUR	2	829	114	6.7	0.2	6.4
15	400 kV	NEEMUCH-Chittorgarh	2	954	94	7.6	0.1	7.5
16	220 kV	BHANPURA-RANPUR	1	0	118	0.0	2.2	-2.2
17	220 kV	BHANPURA-MORAK	1	0	30	0.0	2.2	-2.2
18	220 kV	MEHGAON-AURAIYA	1	104	0	1.5	0.0	1.5
19	220 kV	MALANPUR-AURAIYA	1	75	0	0.9	0.0	0.9
20	132 kV	GWALIOR-SAWAI MADHOPUR	1	0	0	0.0	0.0	0.0
21	132 kV	RAJGHAT-LALITPUR	2	0	0	0.0	0.0	0.0
WR-NR						84.5	216.7	-132.2
Import/Export of WR (With SR)								
1	HVDC	BHADRAWATI B/B	-	0	1011	0.0	20.6	-20.6
2	HVDC	RAIGARH-PUGALUR	2	0	5006	0.0	69.6	-69.6
3	765 kV	SOLAPUR-RAICHUR	2	116	941	0.1	11.2	-11.1
4	765 kV	WARDHA-NIZAMABAD	2	0	1726	0.0	30.1	-30.1
5	765 kV	WARORA-WARANGAL(NEW)	2	0	2343	0.0	42.1	-42.1
6	400 kV	KOLHAPUR-KUDGI	2	1049	0	16.8	0.0	16.8
7	220 kV	KOLHAPUR-CHIKODI	2	0	0	0.0	0.0	0.0
8	220 kV	PONDA-AMBEWADI	1	0	77	1.7	0.0	1.7
9	220 kV	XELDEM-AMBEWADI	1	0	116	2.3	0.0	2.3
WR-SR						20.8	173.5	-152.7

INTERNATIONAL EXCHANGES					Import(+ve)/Export(-ve)		
State	Region	Line Name	Max (MW)	Min (MW)	Avg (MW)	Energy Exchange (MU)	
BHUTAN	ER	400kV MANGDECHHU-ALIPURDUAR 1,2&3 i.e. ALIPURDUAR RECEIPT (from MANGDECHU HEP 4*180MW)	135	-57	6	0.15	
	ER	400kV TALA-BINAGURI 1,2,4 (& 400kV MALBASE - BINAGURI) i.e. BINAGURI RECEIPT (from TALA HEP 6*170MW)	-276	78	-91	-2.18	
	ER	220kV CHUKHA-BIRPARA 1&2 (& 220kV MALBASE - BIRPARA) i.e. BIRPARA RECEIPT (from CHUKHA HEP 4*84MW)	-198	-106	-147	-3.52	
	NER	132kV GELEPHU-SALAKATI	-8	3	-2	-0.05	
	NER	132kV MOTANGA-RANGIA	34	0	33	0.79	
NEPAL	NR	NEPAL IMPORT (FROM UP)	-75	0	-30	-0.71	
	NR	132kV MAHENDRANAGAR-TANAKPUR(NHPC)	-75	0	-50	-1.19	
	ER	NEPAL IMPORT (FROM BIHAR)	-291	0	-23	-0.55	
	ER	400kV DHALKEBAR-MUZAFFARPUR 1&2	-486	-20	-293	-7.04	
BANGLADESH	ER	BHERAMARA B/B HVDC (B'DESH)	-956	-658	-857	-20.57	
	ER (Isolated from Indian Grid)	400kV GODDA_TPS-RAHANPUR (B'DESH) D/C	-1513	-1094	-1237	-29.68	
	NER	132kV COMILLA-SURAJMANI NAGAR 1&2	-138	0	-108	-2.58	

Export From India (in MU)

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.00	0.00	3.78	0.00	0.00	1.29	0.00	0.00	5.07
Nepal	0.24	0.00	5.63	0.00	0.00	0.00	0.00	0.00	5.87
Bangladesh	20.62	0.00	0.00	0.00	0.00	0.00	0.00	0.00	20.62
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Export	20.86	0.00	9.41	0.00	0.00	1.29	0.00	0.00	31.56

Import by India(in MU)

Country	GNA (ISGA/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.33	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.33
Nepal	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Bangladesh	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Import	0.33	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.33

Net from India(in MU)

-ve : Export / +ve : Import

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.33	0.00	-3.78	0.00	0.00	-1.29	0.00	0.00	-4.74
Nepal	-0.24	0.00	-5.63	0.00	0.00	0.00	0.00	0.00	-5.87
Bangladesh	-20.62	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-20.62
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Net	-20.53	0.00	-9.41	0.00	0.00	-1.29	0.00	0.00	-31.23